

Encoding Schemes and Number System

31 August 2026 11:14

'A' = 65 = 0100 0001
↑ ↑ ↑
letter code Binary Number
 (Decimal value)

Encoding of data entered using keyboard.

'3T' on hindi keyboard hex value 0905
whose binary equivalent is 0000 1001 0000 0101

What is encoding?

The mechanism of converting data into an equivalent cipher using specific code is called encoding.

It is important to understand why code value 65 is used for key 'A' and not any other value? It is same for all the keyboards, irrespective of their make?

Yes, it is same for all the keyboards. This has been possible because of standard encoding scheme, where each letter, numerals and symbols is encoded or assigned a unique code. Some of the well-known encoding schemes are:

1. ASCII (American Standard Code for Information Interchange).

7-bit ASCII code.

$2^7 = 128$
@ → 64 a → 97 spacebar → 32 ['a' - 'A']
A → 65
⋮
z → 90 3 → 122 0 → 48, 9 → 57.

! → 33 ' → 34 # → 35 \$ → 36
 % → 37 & → 38 (→ 40) → 41
 ' → 39

UNICODE

Encoding schemes for character sets of different languages. But they were not able to communicate with each other as each of them represents character in their own ways. Hence, text created using one encoding scheme was not recognized by another machine using the different encoding scheme.

So therefore a standard. Called Unicode, has been developed to incorporate all the characters of every written language in of the world. Unicode provides a unique number for every character, irrespective of device (server, desktop, mobile) Operating system (Linux, Windows, IOS) or software applications (different browsers, text editors, etc). Commonly used Unicode encodings are UTF-8, UTF-16 and UTF-32. It is a super set of ASCII and the values zero - 128 have the same character as in ASCII. Unicode characters for Devnagari script is shown in table below.

Devanagari

Showing 120 of 128

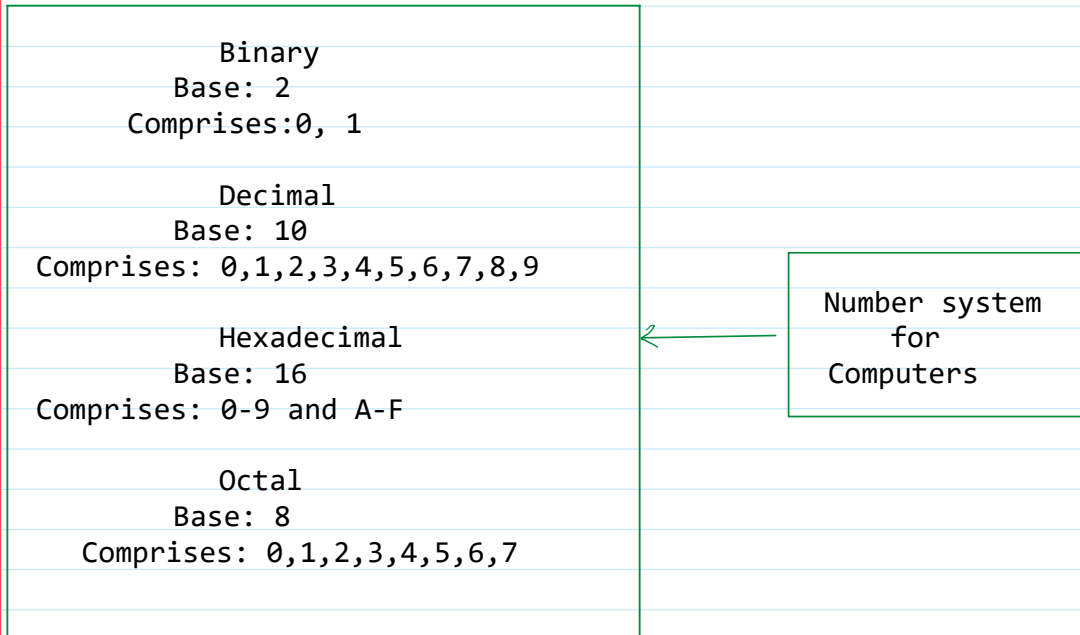
◌̣ U+0900	◌̤ U+0901	◌̥ U+0902	◌̦ U+0903	ओ U+0904	अ U+0905	आ U+0906	इ U+0907
ई U+0908	उ U+0909	ऊ U+090A	ऋ U+090B	लृ U+090C	एँ U+090D	ऐ U+090E	ए U+090F
ऐ U+0910	ऑ U+0911	ओ U+0912	ओ U+0913	औ U+0914	क U+0915	ख U+0916	ग U+0917
घ U+0918	ङ U+0919	च U+091A	छ U+091B	ज U+091C	झ U+091D	ञ U+091E	ट U+091F
ठ U+0920	ड U+0921	ढ U+0922	ण U+0923	त U+0924	थ U+0925	द U+0926	ध U+0927

न U+0928	न U+0929	प U+092A	फ U+092B	ब U+092C	भ U+092D	म U+092E	य U+092F
र U+0930	र U+0931	ल U+0932	ळ U+0933	ळ U+0934	व U+0935	श U+0936	ष U+0937
स U+0938	ह U+0939	ं U+093A	ी U+093B	ः U+093C	ऽ U+093D	ा U+093E	ि U+093F
ी U+0940	ु U+0941	ू U+0942	ृ U+0943	ृ U+0944	ॅ U+0945	ै U+0946	े U+0947
ै U+0948	ॉ U+0949	ो U+094A	ो U+094B	ौ U+094C	् U+094D	ि U+094E	ौ U+094F
ॐ U+0950	ं U+0951	ं U+0952	ं U+0953	ं U+0954	ं U+0955	ं U+0956	ं U+0957
क U+0958	ख U+0959	ग U+095A	ज U+095B	ड U+095C	ढ U+095D	फ़ U+095E	य़ U+095F
ऋ U+0960	ॠ U+0961	ॡ U+0962	ॢ U+0963	। U+0964	॥ U+0965	० U+0966	१ U+0967
२ U+0968	३ U+0969	४ U+096A	५ U+096B	६ U+096C	७ U+096D	८ U+096E	९ U+096F
० U+0970	१ U+0971	अ U+0972	अ U+0973	आ U+0974	औ U+0975	अ U+0976	अ U+0977
रु U+0978	ज़ U+0979	ष U+097A	ग U+097B	ज U+097C	१ U+097D	ड U+097E	ब U+097F

Number System

Every number system has a set of unique characters or literals. The count

of these literals is called the radix or base of the number system. The four different number system used in this context of computer.



$$\begin{array}{r} 123.45 \\ - 19 \\ \hline 02 \end{array}$$

Four different number systems.

Integer part fractional part ← real numbers

Digit	1	2	3	.	4	5
Positional number	2	1	0		-1	-2
Position value	10^2	10^1	10^0		10^{-1}	10^{-2}

$$1 \times 10^2 + 2 \times 10^1 + 3 \times 10^0 + 4 \times 1/10 + 5 \times 1/100 \\ 100 + 20 + 3 + 0.25 + 0.05 = (123.45)_{10}$$

Computation of decimal number using its positional value.

Binary Number System

Digit	1	0	1	1	.	1	0	1
Positional number	3	2	1	0		-1	-2	-3
Positional value	2^3	2^2	2^1	2^0		2^{-1}	2^{-2}	2^{-3}

$$1 * 8 + 1 * 2 + 1 * 1 + 1/2 + 1/8 = 11.625$$

Octal number system: Octal number system is called base - 8 system. As its total digits (0 - 7) And positional value is expressed in powers of 8.

Digit	2	4	7	.	2	4
Positional number	2	1	0		-1	-2
Positional value	8^2	8^1	8^0		8^{-1}	8^{-2}

$$2 * 8^2 + 4 * 8 + 7 * 1 + 2 * 1/8 + 4/64 = 167.3125 \text{ (This is decimal number)}$$

Hexadecimal number system:

Hexadecimal numbers are also used for compact representation of binary numbers. It consists of 16 unique symbols (0 - 9, A - F) and is called the base - 16 system. In hexadecimal system, each alphanumeric digit is represented as a group of four binary digits because four bits ($2^4 = 16$) Are sufficient to represent 16 alphanumeric symbols.

Digit	2	3	A	.	0	5
Positional number	2	1	0		-1	-2
Positional value	256	16	1		1/16	1/256

$$256*2 + 16 * 3 + 10 * 1 + 5/256 = 570.0195 \text{ (Decimal Value)}$$

Applications of hexadecimal number system.

- Main memory is made up of memory locations where each location has a unique address. Usually size of a memory address is 16 bit or 32 bit. To access 16 bit memory address, a programmer has to use 16 binary bits, which is difficult to deal with. To simplify the address representation, hexadecimal and octal numbers are used. Let us consider for a 16 bit memory address 1100000011110001. Using the hexadecimal notation, this address is mapped to C0F1. Which is more easy to remember. The equivalent octal representation for 16 - bit value is **140361**.

Python code:

```
print(oct(0b1100000011110001))
0o140361
```

- Hexadecimal numbers are also used for describing the colors on the web page. Each color is made up of three primary colors, red, green and blue, popularly called RGB, in short. In most color maps, each color is usually chosen from the palette of 16 million colors. Therefore, 24 bits are required for representing each color having three components. (8 bits were read, 8 bits for green, eight bits for blue component) It is difficult to remember 24 - bit binary color code. Therefore, color codes are written in hexadecimal form for compact representation. For example 24 - bit code for red color is. 1 1 1 1 1 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0. The equivalent hexadecimal notation is (FF,00,00), which can be easily remembered.

Conversion between number system

Conversion from decimal to other number systems

To convert a decimal number to any other number system (binary optal or

hexadecimal) use the steps given below.

1. Divide the given number by the base value (B) of the number system in which it is to be converted.
2. Note the remainder.
3. Keep on dividing the quotient by the base value and note the remainder till the quotient is zero.
4. Write the noted remainders in the reverse order (from bottom to top).

Decimal to Binary Conversion

2	65	
2	32	1
2	16	0
2	8	0
2	4	0
2	2	0
2	1	0
	0	1

$$(65)_{10} = (1000001)_2$$

2	122	
2	61	0
2	30	1
2	15	0
2	7	1
2	3	1
2	1	1
	0	1

$$(122)_{10} = (1111010)_2$$

$$(125.55)_{10} = (\quad ? \quad)_2$$

2	125	
2	62	1
2	31	0
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

	55 × 2
1	10 × 2
0	20 × 2
0	40 × 2
0	80 × 2
1	60 × 2
1	20

$$(125.55)_{10} \approx (111101.100011)_2$$

Decimal to octal.

$$(65)_{10} = (?)_8$$

$$\begin{array}{r|l} 8 & 65 \\ \hline 8 & 8 \\ \hline 8 & 1 \\ \hline & 0 \end{array}$$

$$\begin{array}{c} 1 \\ 0 \\ 1 \end{array} \uparrow$$

$$(65)_{10} = (101)_8$$

$$(122)_{10} = (?)_8$$

$$\begin{array}{r|l} 8 & 122 \\ \hline 8 & 15 \\ \hline 8 & 1 \\ \hline & 0 \end{array}$$

$$\begin{array}{c} 2 \\ 7 \\ 1 \end{array} \uparrow$$

$$(122)_{10} = (172)_8$$

$$(65.45)_{10} = (?)_8$$

$$\begin{array}{r|l} 8 & 65 \\ \hline 8 & 8 \\ \hline 8 & 1 \\ \hline & 0 \end{array}$$

$$\begin{array}{c} 1 \\ 0 \\ 1 \end{array} \uparrow$$

$$\begin{array}{r|l} 3 & 45 \times 8 \\ \hline 3 & 60 \times 8 \\ \hline 4 & 80 \times 8 \\ \hline 6 & 40 \times 8 \\ \hline 3 & 20 \times 8 \\ \hline 1 & 60 \end{array}$$

$$(65.45)_{10} \approx (101.34631)_8$$

Decimal to hexadecimal

$$(65)_{10} = (?)_{16}$$

$$\begin{array}{r|l} 16 & 65 \\ \hline 16 & 4 \\ \hline & 0 \end{array}$$

$$\begin{array}{c} 1 \\ 4 \end{array} \uparrow$$

$$(65)_{10} = (41)_{16}$$

$$(752)_{10} = (?)_{16}$$

$$\begin{array}{r|l} 16 & 752 \\ \hline 16 & 47 \\ \hline 16 & 3 \\ \hline & 0 \end{array}$$

$$\begin{array}{c} 0 \\ 1 \\ 3 \end{array} \uparrow$$

$$(752)_{10} = (310)_{16}$$

$$(933)_{10} = (?)_{16}$$

$$(933)_{10} = (?)_{16}$$

16	933
16	58
16	3
	0

5	↑
10	A
3	

$$(933)_{10} = (3A5)_{16}$$

$$(933.83)_{10} = (?)_{16}$$

$$\Rightarrow (933)_{10} = (3A5)_{16}$$

		4
		83 × 16
	12	28 × 16
4	12	48 × 16
7	12	68 × 16
		88 ×

D ← 13
A ← 10

$$(933.83)_{10} \approx (3A5.D47A)_{16}$$

Decimal → Binary
 Decimal → Octal
 Decimal → Hexadecimal.

Binary → Decimal
 Binary → Octal
 Binary → Hexadecimal

Binary to Decimal.

$$(1101011)_2 = (?)_{10}$$

6	5	4	3	2	1	0
1	1	0	1	0	1	1

$$= 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$= 64 + 32 + 8 + 2 + 1 = 107_{10}$$

$$(1101011.010111)_2$$

$$= 1 \times 2^6 + 1 \times 2^5 + 0 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 + 0 \times 2^{-1} + 1 \times 2^{-2} + 0 \times 2^{-3} + 1 \times 2^{-4}$$

$$= 64 + 32 + 8 + 2 + 1 + \frac{1}{4} + \frac{1}{16}$$

$$= 107 + \frac{4+1}{16} = 107 + 0.3125 = 107.3125 \text{ Ans}$$

8	4	2	1
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$$0000 = 0$$

$$0001 = 1$$

$$+ \dots = \dots$$

0.3125
16 50
48
20
16
40
32
80
80
0

0	0	0	0	= 0
0	0	0	1	= 1
0	0	1	0	= 2
0	0	1	1	= 3
0	1	0	0	= 4
0	1	0	1	= 5
0	1	1	0	= 6
0	1	1	1	= 7
1	0	0	0	= 8
1	0	0	1	= 9
1	0	1	0	= 10

11	10	9	8	7	6	5	4	3	2	1	0
2048	1024	512	256	128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	1	1	0	1	1
0	0	0	0	0	1	0	1	0	0	0	1

$2048+1024+512+256+128+64+32+16+8+4+2+1 = 4095$

$$2048 + 1024 = 3072$$

$$(163)_{10} =$$

1101101

11	10	9	8	7	6	5	4	3	2	1	0
2048	1024	512	256	128	64	32	16	8	4	2	1
1	1	0	1	1	0	1					

$$64 + 32 + 8 + 4 + 1 = 109$$

Binary to octal. ($8 = 2^3$) ← That's why 3-bit

1. Create a group of 3-bits from right to left
2. Convert each group of 3-bits into digits.
3. The number you get is in octal.

4	2	1
1	1	1
0	1	0
1	0	1

= 7
= 2
= 5

[Note: add leading zero to create the group. OR, add trailing zero after decimal point value to create the group.]

$$(10101011)_2 = (527)_8$$

$$(001100101)_2 = (145)_8$$

$$(001100101.101100)_2 = (145.54)_8$$

Octal to Binary

Octal to Binary

1. Convert each digit into 3-bit binary value.
2. concatenate all the values. That will be the final answer.

$$(527)_8 = (101010111)_2$$

$$(145)_8 = (001100101)_2$$

$$(145.54) = (1100101.101100)_2$$

Binary to hexadecimal $16 = 2^4$ → That's why 4-bits

1. Create a group of 4-bits from right to left
2. Convert each group of 4-bits into digits.
3. The number you get is in hexadecimal.

4	2	1
---	---	---

$$\begin{array}{r} 111 = 7 \\ 010 = 2 \\ 101 = 5 \end{array}$$

[Note: add leading zero to create the group. OR, add trailing zero after decimal point value to create the group.]

$$(1010101011)_2 = (AAB)_{16}$$

$$(000101001111)_2 = (14F)_{16}$$

$$(00011011.10101000)_2 = (1B.A8)_{16}$$

8	4	2	1
---	---	---	---

$$\begin{array}{r} 1010 = 10 \\ 1011 = 11 \\ 1111 = 15 \\ 0100 = 4 \\ 0001 = 1 \\ 1000 = 8 \end{array}$$

Hexadecimal to binary.

1. Convert each digit into 4-bit binary value.
2. concatenate all the values. That will be the final answer.

$$(AAB)_{16} = (1010101011)_2$$

$$(14F)_{16} = (101001111)_2$$

$$(1B.A8) = (11011.10101000)_2$$

Octal to Decimal

$$(761)_8 = ()_{10}$$

$$761$$

$$64 \times 7$$

$$(761)_8 = (\quad)_{10}$$

$$\begin{array}{l} 761 \\ \begin{array}{l} \rightarrow 1 \times 8^0 = \\ \rightarrow 6 \times 8^1 = \\ \rightarrow 7 \times 8^2 = \end{array} \end{array} \quad \begin{array}{r} 1 \\ 48 \\ 448 \\ \hline 497 \end{array}_{10}$$

$$(761)_8 = (497)_{10}$$

$$(761.25)_8 = (\quad)_{10}$$

$$7 \times 8^2 + 6 \times 8^1 + 1 \times 8^0 + 2 \times 8^{-1} + 5 \times 8^{-2}$$

$$497 + \frac{2}{8} + \frac{5}{16}$$

$$\begin{array}{r} 4 + 5 \\ \hline 16 \end{array}$$

$$497 + 0.5625$$

$$= (497.5625)_{10}$$

$$2 \times 64 \times 7$$

$$\begin{array}{r} 0.5625 \\ 16 \overline{) 90} \\ \underline{80} \\ 100 \\ \underline{96} \\ 40 \\ \underline{32} \\ 80 \end{array}$$

Octal to hexadecimal

1. Convert each octal digit into 3-bit binary values.
2. Create a group of 4-bit binary from right to left in integer part and, left to right after decimal point. Add leading/trailing zeros as per requirement.
3. The value you get is hexadecimal value.

$$(765)_8 = (\quad ? \quad)_{16}$$

$$(00011110101) = (1F5)_{16}$$

8	4	2	1
0	1	0	1

$$1111 = 15 \rightarrow F$$

$$(607.25)_8 = (\quad)_{16}$$

$$(000110000111.01010100) = (187.54)_{16}$$

Hexadecimal to octal

1. Convert each hexadecimal digit into 4-bit binary values.
2. Create a group of 3-bit binary from right to left in integer part and, left to right after decimal point. Add leading/trailing zeros as per requirement.

3. The value you get is octal value.

zeros as per requirement.

3. The value you get is octal value.

$$(1F5)_{16} = (?)_8$$

7	2	1
10	1	
11	0	

$$(11110101) = (765)_8$$

$$(127.54)_{16} = (?)_8$$

$$(11000011).010101000) = (607.25)_8$$

Homework:

EXERCISE

- Write base values of binary, octal and hexadecimal number system.
- Give full form of ASCII and ISCII.
- Try the following conversions.
 - $(514)_8 = (?)_{10}$
 - $(220)_8 = (?)_2$
 - $(76F)_{16} = (?)_{10}$
 - $(4D9)_{16} = (?)_{10}$
 - $(11001010)_2 = (?)_{10}$
 - $(1010111)_2 = (?)_{10}$
- Do the following conversions from decimal number to other number systems.
 - $(54)_{10} = (?)_2$
 - $(120)_{10} = (?)_2$
 - $(76)_{10} = (?)_8$
 - $(889)_{10} = (?)_8$
 - $(789)_{10} = (?)_{16}$
 - $(108)_{10} = (?)_{16}$
- Express the following octal numbers into their equivalent decimal numbers.
 - 145
 - 6760
 - 455
 - 10.75

6. Express the following decimal numbers into hexadecimal numbers.
(i) 548 (ii) 4052 (iii) 58 (iv) 100.25
7. Express the following hexadecimal numbers into equivalent decimal numbers.
(i) 4A2 (ii) 9E1A (iii) 6BD (iv) 6C.34
8. Convert the following binary numbers into octal and hexadecimal numbers.
(i) 1110001000 (ii) 110110101 (iii) 1010100
(iv) 1010.1001
9. Write binary equivalent of the following octal numbers.
(i) 2306 (ii) 5610 (iii) 742 (iv) 65.203
10. Write binary representation of the following hexadecimal numbers.
(i) 4026 (ii) BCA1 (iii) 98E (iv) 132.45
11. How does computer understand the following text?
(hint: 7 bit ASCII code).
(i) HOTS (ii) Main (iii) CaSe
12. The hexadecimal number system uses 16 literals (0–9, A–F). Write down its base value.
13. Let X be a number system having B symbols only. Write down the base value of this number system.
14. Write the equivalent hexadecimal and binary values for each character of the phrase given below.
“ हम सब एक ”
15. What is the advantage of preparing a digital content in Indian language using UNICODE font?
16. Explore and list the steps required to type in an Indian language using UNICODE.
17. Encode the word ‘COMPUTER’ using ASCII and convert the encode value into binary values.